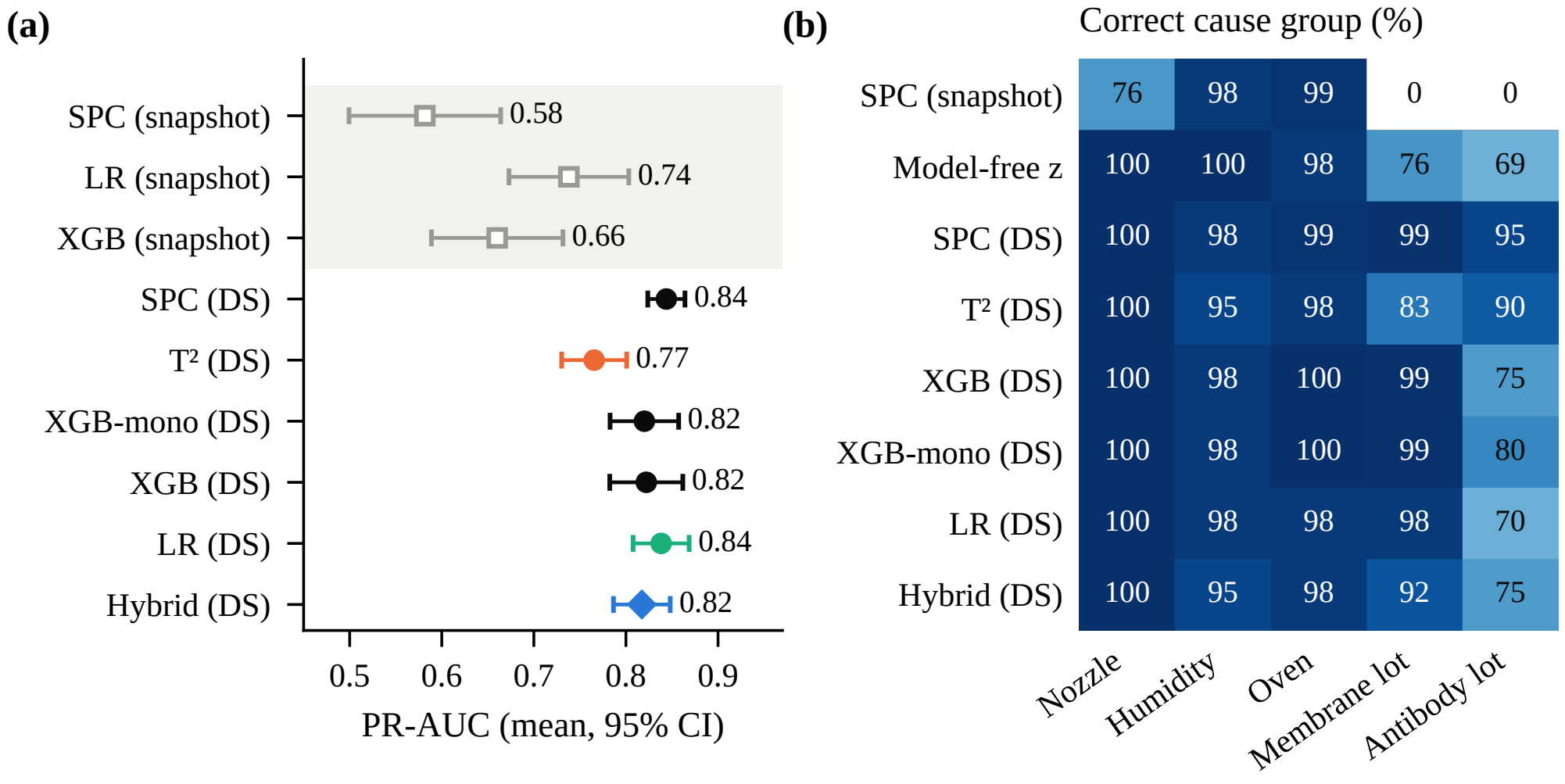
(a)
SPC (snapshot)
LR (snapshot)
XGB (snapshot)
SPC (DS)
T² (DS)
XGB-mono (DS)
XGB (DS)
LR (DS)
Hybrid (DS)
0.58
0.74
0.66
0.84
0.77
0.82
0.82
0.84
0.82
0.5
0.6
0.7
0.8
0.9
PR-AUC (mean, 95% CI)
(b)
Correct cause group (%)
SPC (snapshot) 76 98 99 0 0
Model-free z 100 100 98 76 69
SPC (DS) 100 98 99 99 95
T² (DS) 100 95 98 83 90
XGB (DS) 100 98 100 99 75
XGB-mono (DS) 100 98 100 99 80
LR (DS) 100 98 98 98 70
Hybrid (DS) 100 95 98 92 75
Nozzle
Humidity
Oven
Membrane lot
Antibody lot